

The self-sufficiency contrast: where Newton's axioms leave motion undefined, the theory with $h > 0$ defines it

The thesis in STATE says that Newton's mechanics is the large-action regime of a theory with $h > 0$, and that it needs h because its own axioms do not make a complete theory of motion. This note proves the first piece of evidence for that claim as one theorem with four parts, for Kepler's force. The classical flow of $V = -k/r$ is incomplete: every zero-angular-momentum datum that moves inward, or is bound, reaches the centre in finite time, with the explicit collision time $t_c = \frac{\pi}{2} \sqrt{mr_0^3/2k}$ for fall from rest, and Newton's axioms give no continuation there. The quantum Hamiltonian $-\frac{h^2}{2m}\Delta - k/r$ is self-adjoint on the domain of the Laplacian and generates a global unitary group for every state and every $h > 0$, by Kato's theorem, cited with its hypotheses. The same contrast holds for the quartic between the fixed- h unitary limit and the classical-first construction that needs a branch rule. And the classical trajectory is a derived object of the $h > 0$ theory: for quadratic forces the centroid of any state obeys Newton's equation exactly at every h , and for Kepler's force a coherent state follows the orbit in the large-action regime for finite times away from the collision set, by Hepp's theorem, cited with its hypotheses and with the one further localization hypothesis the singular potential needs. Parts (a), (c) and the quadratic half of (d) are proved here; (b) and the Kepler half of (d) rest on the cited theorems. Proposition 2 then gives the gap the exercise is for: with finite c , the theories of Kepler's force that define motion without an added rule are exactly those with $h \geq k/(\alpha_c c)$, by the self-adjointness thresholds of the relativistic Coulomb problem, cited; Newton's theory at $h = 0$ and every theory with $0 < h < k/(\alpha_c c)$ are not self-sufficient. Written by Claude Fable, 2026-10-01; proof status recorded in the final section.

Notation: h is the reduced phase constant, $m > 0$ the mass, $k > 0$ the force constant, $r = |x|$ in $\mathbb{R}^3$; the planar case is a remark.

1. Statement

Theorem 1 (self-sufficiency contrast for Kepler's force).

- (a) *Classical incompleteness.* Let $L = x \times p$ be the angular momentum. Every initial datum with $L = 0$ whose radial velocity is inward or zero, and every datum with $L = 0$ and negative energy, reaches $r = 0$ at a finite time. For a body released from rest at distance r_0 the collision time is

$$t_c = \frac{\pi}{2} \sqrt{\frac{mr_0^3}{2k}}, \quad (1)$$

and in general $t_c \leq \frac{\pi}{2} \sqrt{mr_{\max}^3/2k}$ with $r_{\max}$ the largest distance reached. The collision set $\mathcal{C} = \{L = 0, \dot{r} \leq 0 \text{ or } E < 0\}$ is nonempty, invariant under the flow while it exists, and of measure zero in phase space. On $\mathcal{C}$ the vector field is unbounded at the endpoint and Newton's axioms, the Laws together with the force law, define no continuation: the flow of Kepler's force is not complete.

- (b) *Quantum completeness.* For every $h > 0$, $m > 0$ and real k , the operator

$$H_h = -\frac{h^2}{2m}\Delta - \frac{k}{r} \quad \text{on } L^2(\mathbb{R}^3) \quad (2)$$

is self-adjoint on the domain $H^2(\mathbb{R}^3)$ of the Laplacian and bounded below, and $U_h(t) = e^{-itH_h/h}$ is a strongly continuous unitary group defined for every $t \in \mathbb{R}$ and every initial state in $L^2(\mathbb{R}^3)$. This is Kato's theorem [Kato1951; metadata], whose hypothesis, that the potential is the sum of an $L^2(\mathbb{R}^3)$ and a bounded function, is verified in §3.

- (c) *The quartic contrast.* For $V = \lambda q^4$ on the line, Theorem 2(b) of the 1998-conjecture note gives, for every $h > 0$, the strong convergence of the time-sliced operators to the unitary evolution $e^{-iTH_h/h}$ along every sequence of partitions, with no extra datum; Theorem 1 there shows that the halved classical-first construction at zero resolution has no limit on a two-cell partition with three stationary paths, and Theorem 2(a) there repairs it only by a branch rule chosen by hand. The fixed- h construction exists for all data; the construction at zero needs a rule the action does not supply.
- (d) *The classical trajectory as a derived object.* (d1) For $H = p^2/2m + V$ with $\deg V \leq 2$, the expectations $\langle q \rangle_t, \langle p \rangle_t$ of every state with finite second moments obey Newton's equations exactly for every $h > 0$,

$$\frac{d}{dt}\langle q \rangle_t = \frac{\langle p \rangle_t}{m}, \quad \frac{d}{dt}\langle p \rangle_t = -V'(\langle q \rangle_t), \quad (3)$$

and a coherent state stays a Gaussian whose centroid is the classical trajectory. (d2) For Kepler's force, let $z_0 = (x_0, p_0)$ lie on a classical orbit whose distance from the centre stays at least $\rho_{\min} > 0$ on $[0, T]$. Under Hepp's theorem, stated in §5 with its hypotheses, applied to a smooth potential V_δ equal to $-k/r$ for $r \geq \delta$ with $0 < \delta < \rho_{\min}$, the coherent state $\varphi_{z_0}^h$ with $\sigma_h^2 \propto h$ evolves under $H_h^{(\delta)}$ into a Gaussian centred on the Kepler orbit z_t up to an L^2 error tending to zero as $h \rightarrow 0$, uniformly on $[0, T]$; and under the localization hypothesis (L) of §5 the same holds for the Coulomb evolution $U_h(t)$ itself.

Conclusion. The theory with $h > 0$ defines motion for all data and all times, by (b), and yields Newton's trajectories where they exist, by (d), while Newton's axioms leave motion undefined on the collision set, by (a), and the construction at zero needs a rule the axioms do not contain, by (c).

2. Proof of (a)

Conservation of L under a central force is Proposition I of Book I; for $L = 0$ the motion is on a line through the centre and $E = \frac{m}{2}\dot{r}^2 - k/r$ is conserved. Released from rest at r_0 , $E = -k/r_0$ and

$$\dot{r}^2 = \frac{2k}{m} \left(\frac{1}{r} - \frac{1}{r_0} \right) = \frac{2k}{m} \frac{r_0 - r}{r r_0},$$

with $\dot{r} < 0$ for $t > 0$ because the force points inward. Hence

$$t_c = \int_0^{r_0} \frac{dr}{|\dot{r}|} = \sqrt{\frac{mr_0}{2k}} \int_0^{r_0} \sqrt{\frac{r}{r_0 - r}} dr.$$

With $r = r_0 \sin^2 \theta$, $dr = 2r_0 \sin \theta \cos \theta d\theta$ and $\sqrt{r/(r_0 - r)} = \tan \theta$, the integral is $2r_0 \int_0^{\pi/2} \sin^2 \theta d\theta = \pi r_0/2$, which gives (1).

For a general datum with $L = 0$ and $\dot{r} \leq 0$: if $E < 0$ then $r \leq r_{\max} = -k/E$ throughout, $|\dot{r}|^2 = \frac{2k}{m}(1/r - 1/r_{\max})$, and the time to the centre is at most the fall time from rest at $r_{\max}$, which is (1) with $r_{\max}$; if $E \geq 0$ then $|\dot{r}| \geq \sqrt{2k/(mr)}$ and the time is at most $\int_0^{r_0} \sqrt{mr/2k} dr = \frac{2}{3} \sqrt{m/2k} r_0^{3/2}$. If $L = 0$, $E < 0$ and $\dot{r} > 0$, the body reaches $r_{\max}$ at a finite time and then falls as before. Invariance of $\mathcal{C}$ follows from the conservation of L and E ; $L = 0$ is the condition that p be parallel to x , two independent equations on the six-dimensional phase space, so $\mathcal{C}$ has measure zero. At $r = 0$ the force $-kx/r^3$ is unbounded; no existence or uniqueness theorem applies there, and the Laws, which presuppose a defined force, assign no motion. A continuation through the centre (Levi-Civita or Kustaanheimo–Stiefel regularization) is an added rule, in the same position as the branch rule of (c). $\square$

3. Proof of (b), with Kato's hypothesis verified

Kato's theorem [Kato1951; metadata; the statement used is the one standardly quoted from it, the primary text was not read in this session]: let $V = V_1 + V_2$ be real with $V_1 \in L^2(\mathbb{R}^3)$ and $V_2 \in L^\infty(\mathbb{R}^3)$. Then for every $a > 0$ the operator $-a\Delta + V$ is self-adjoint on $H^2(\mathbb{R}^3)$ and bounded below. The mechanism is the Sobolev bound $\|u\|_\infty \leq \epsilon\|\Delta u\| + C_\epsilon\|u\|$ for $u \in H^2(\mathbb{R}^3)$, which makes V_1 relatively bounded with respect to Δ with relative bound zero, so that the Kato–Rellich theorem applies for every value of $a = h^2/2m$.

Verification for $V = -k/r$: put $V_1 = -\frac{k}{r}\mathbf{1}_{r<1}$ and $V_2 = -\frac{k}{r}\mathbf{1}_{r\geq 1}$. Then $|V_2| \leq |k|$, and

$$\int_{\mathbb{R}^3} |V_1|^2 d^3x = k^2 \int_0^1 \frac{4\pi r^2}{r^2} dr = 4\pi k^2 < \infty.$$

So the hypothesis holds for every real k , and H_h in (2) is self-adjoint on $H^2(\mathbb{R}^3)$ for every $h > 0$, $m > 0$. Stone's theorem then gives the unitary group $U_h(t)$ for all $t \in \mathbb{R}$, strongly continuous, defined on every $\psi \in L^2$, with $U_h(t)\psi \in H^2$ for $\psi \in H^2$. No initial datum and no time is excluded; the states supported near $r = 0$ evolve like all others. $\square$

Planar remark. In $\mathbb{R}^2$ the function $1/r$ is not square integrable near the origin, and the operator-sense theorem does not apply; the planar Coulomb Hamiltonian is defined in the form sense by the KLMN theorem, since $1/r \in L^p_{\text{loc}}(\mathbb{R}^2)$ for $p < 2$ is form-bounded relative to $-\Delta$ with relative bound zero. The planar Kepler problem used elsewhere in this repository therefore also has a global unitary dynamics, by that standard route, which is not reproved here.

4. Proof of (c)

The statements are Theorems 1, 2(a) and 2(b) of the 1998-conjecture note, accepted on refereeing there (status line of that note). Theorem 2(b) proves the product formula for $H_h = -\frac{h^2}{2m}d^2/dq^2 + \lambda q^4$ at every $h > 0$: essential self-adjointness, the strong limit of the symmetric time slicing along any partition sequence, and the composition $\|U_h(T)\| = 1$. Theorem 1 proves that the halved classical-first functional on the two-cell partition has, as the resolution tends to zero, the oscillating value (4) there with no limit, because the three stationary paths interfere; Theorem 2(a) restores a limit only by restricting to a neighbourhood of one path, a datum the action does not contain. Hence for the quartic the $h > 0$ theory exists for all data without extra rules and the construction at zero does not. $\square$

5. Proof of (d)

(d1), *quadratic forces*. By Theorem A(a) of the fifth-postulate note, the Heisenberg equations are Newton's second law as operator identities for every real h : $\dot{q} = p/m$ and $\dot{p} = -V'(q)$. Taking expectations in a state with finite second moments gives $\frac{d}{dt}\langle q \rangle = \langle p \rangle/m$ and $\frac{d}{dt}\langle p \rangle = -\langle V'(q) \rangle$; for $\deg V \leq 2$ the function V' is affine, so $\langle V'(q) \rangle = V'(\langle q \rangle)$, which is (3). The centroid therefore follows the classical trajectory through $(\langle q \rangle_0, \langle p \rangle_0)$ for every $h > 0$ and every such state, with no approximation. For a coherent state the Wigner function is the Gaussian transported by the affine flow (proof of Theorem 5 of the reachability note), so the state stays Gaussian with the classical trajectory as its centroid and covariance $M_t \Sigma_h M_t^\top$; in the large-action regime of §3c there, masses scaled by λ at fixed h and widths, the position width $s_t(h)$ tends to the initial σ and the state is as localized on its trajectory as it was prepared. $\square$

(d2), *Kepler's force*. Hepp's theorem [Hepp1974; metadata; the statement used is the one standardly quoted from it, the primary text was not read in this session], in the form used here: let W be a real C^∞ potential on $\mathbb{R}^3$ with bounded derivatives of every order ≥ 2 , so that $-\frac{h^2}{2m}\Delta + W$ is self-adjoint and the classical flow Φ_t of $p^2/2m + W$ is complete; let φ_z^h be the coherent state at $z = (x, p)$ with position width σ_h , $\sigma_h^2 = hL_*/P_*$ for fixed L_*, P_* ; and let

φ_{z_t, M_t}^h be the Gaussian centred at $z_t = \Phi_t(z_0)$ with covariance transported by the linearized flow $M_t = D\Phi_t(z_0)$. Then for every $T > 0$,

$$\sup_{|t| \leq T} \|e^{-itH_h^W/h} \varphi_{z_0}^h - e^{i\theta_t/h} \varphi_{z_t, M_t}^h\|_{L^2} \longrightarrow 0 \quad (h \rightarrow 0), \quad (4)$$

with θ_t the classical action along the trajectory. Apply this with $W = V_\delta$, a C^∞ function equal to $-k/r$ for $r \geq \delta$ and bounded with bounded derivatives inside the ball: for $r \geq \delta$ all derivatives of $-k/r$ of order ≥ 1 are bounded by constants times $\delta^{-2}, \delta^{-3}, \dots$, so V_δ satisfies the hypothesis. Since the Kepler orbit through z_0 stays at distance $\geq \rho_{\min} > \delta$ on $[0, T]$, it is also the orbit of V_δ , and (4) gives the first claim of (d2): under $H_h^{(\delta)}$ the coherent state follows the Kepler orbit.

For the Coulomb evolution itself, Duhamel's formula on the common domain gives

$$U_h(t)\varphi - U_h^{(\delta)}(t)\varphi = -\frac{i}{h} \int_0^t U_h(t-s) (V - V_\delta) U_h^{(\delta)}(s)\varphi ds,$$

so that $\|U_h(t)\varphi - U_h^{(\delta)}(t)\varphi\| \leq \frac{T}{h} \sup_{s \leq T} \|(V - V_\delta)U_h^{(\delta)}(s)\varphi\|$, where $V - V_\delta$ is supported in the ball $r < \delta$. **Hypothesis (L):** $\sup_{s \leq T} \|(V - V_\delta)U_h^{(\delta)}(s)\varphi_{z_0}^h\| = o(h)$ as $h \rightarrow 0$. Under (L) the Coulomb and regularized evolutions of the coherent state coincide in the limit and (4) transfers to U_h . The hypothesis says that the evolved state carries negligible mass and energy inside the ball the orbit never visits; for the Gaussian φ_{z_t, M_t}^h alone it holds with an exponentially small bound, by Hardy's inequality $\|f/r\| \leq 2\|\nabla f\|$ in $\mathbb{R}^3$ and the Gaussian tails, but the $o(1)$ remainder in (4) is an L^2 statement, and (L) requires its energy-norm counterpart, which is not proved here. $\square$

5b. Proposition 2: with finite c , self-sufficiency has a smallest h

The point of the exercise is a gap in the admissible values of h , the elementary analogue of a spectrum $\{0\} \cup [m, \infty)$. Theorem 1 alone gives none: by (b) every $h > 0$ makes the non-relativistic Coulomb theory complete, and the ground-state energy $-mk^2/2h^2$ tends to $-\infty$ as $h \rightarrow 0$ without any threshold. The threshold appears when the speed of light is finite, through the known self-adjointness limits of the relativistic Coulomb problem.

Proposition 2 (smallest h at finite c). Let $c < \infty$ and $\nu = k/(hc)$, the Coulomb coupling in units of hc . (i) *Spin one half.* The Dirac–Coulomb operator $c\alpha \cdot p + \beta mc^2 - k/r$ on $L^2(\mathbb{R}^3)^4$ has a distinguished self-adjoint extension, selected by finiteness of the potential energy on its domain, for $\nu < 1$ [Schmincke1972; Wuest1975; Nenciu1976; metadata], and is essentially self-adjoint on $C_c^\infty(\mathbb{R}^3 \setminus \{0\})^4$ for $\nu \leq \sqrt{3}/2$; for $\nu > 1$ every self-adjoint realization requires a boundary condition at the centre, an added rule in the sense of Theorem 1(a) and (c). (ii) *Spin zero.* For the Klein–Gordon equation with the Coulomb potential the s -wave radial problem is regular at the origin iff $\nu \leq \frac{1}{2}$ [Case1950; metadata]; beyond it the solutions oscillate without limit at the centre and a boundary condition must be imposed. (iii) Hence the theories of Kepler's force at finite c that define the motion without an added rule are exactly those with

$$h \geq h_{\min} = \frac{k}{\alpha_c c}, \quad \alpha_c = 1 \text{ (spin } \frac{1}{2}), \quad \alpha_c = \frac{1}{2} \text{ (spin } 0), \quad (5)$$

with the strict inequality in the spin-one-half case. Together with Theorem 1(a), the set of values of h at which a theory of Kepler's force is self-sufficient is $[h_{\min}, \infty)$: Newton's theory at $h = 0$ is incomplete, no theory with $0 < h < h_{\min}$ is self-sufficient, and every theory with $h \geq h_{\min}$ is. This is the gap. In the non-relativistic limit $c \rightarrow \infty$, $h_{\min} \rightarrow 0$ and the gap closes, in agreement with Theorem 1(b).

Proof. (i) and (ii) are the cited theorems, read at the metadata level and in their standard statements; the thresholds are stated as they are standardly quoted, with the essential self-adjointness bound $\sqrt{3}/2$ due to Weidmann, not separately cited here. (iii) is the translation $\nu < \alpha_c \iff h > k/(\alpha_c c)$, together with Theorem 1(a) at $h = 0$. $\square$

The unit k/c is the one the necessity-unit note isolated as the unique mass-independent action built from k_e and c , and the relativistic Kepler note found the same threshold classically as $\ell > k/c$ for regular bound motion. Proposition 2 places it where the thesis needs it: as the lower edge of the set of self-sufficient theories. What it does not do is fix h : any $h \geq h_{\min}$ is admissible, and the physical value exceeds $h_{\min}$ by the inverse fine-structure ratio, which is the separate calibration obligation.

6. What the theorem establishes and what it does not

The contrast is between two theories of the same force law. Newton's, the Laws with $F = -kx/r^3$, assigns no motion to the collision set and, for the quartic, cannot even be constructed at zero resolution without a rule the action does not supply. The theory with $h > 0$ assigns a motion to every state for all time, by self-adjointness, and recovers Newton's trajectories as the centroids of localized states, exactly for quadratic forces and, in the large-action regime and away from collisions, for Kepler's force under the cited theorem. The thesis in STATE reads this as the sense in which Newton's mechanics needs h : its only completion is the large-action regime of a theory that contains h .

Not established here: the energy-norm localization (L) for the Coulomb evolution; the planar operator beyond the standard form construction; any statement that no completion of Newton's mechanics without h exists, which is the open obligation (ii) of STATE; and anything about long times, where the Ehrenfest-type breakdown of (4) is a separate matter. The theorem is evidence for the thesis, in the form the thesis now has, and no more.

7. Consequence for STATE

Proposition 2 supplies the gap: the self-sufficient theories of Kepler's force at finite c have $h \geq k/(\alpha_c c)$, with Newton at $h = 0$ outside; it rests on cited thresholds and fixes no value of h . Theorem 1 supplies the self-sufficiency contrast required by STATE's obligation (iii): classical incompleteness with the explicit collision time and collision set (proved), quantum completeness by Kato's theorem (cited, hypothesis verified), the quartic contrast (proved in the 1998-conjecture note), and the classical trajectory as the derived centroid, exact for quadratic forces (proved) and for Kepler's force under Hepp's theorem and hypothesis (L) (cited). Next: prove (L) for the Coulomb evolution of a coherent state, or replace Hepp's L^2 statement by an energy-norm one; then the open obligation (ii), that no completion without h exists. Proof status: written by Claude Fable, 2026-10-01; parts (a), (c), (d1) proved, (b) and (d2) cited with hypotheses; refereeing recorded in the status line when done.